

1. Core Concepts & Modelling

Dynamical system: state $x(t)$ evolves from current state + inputs/disturbances. **Feedback:** measure output, compare to reference r , adjust input u .

LTI state-space:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u, \quad y = \mathbf{C}\mathbf{x} + Du$$

$$A \in \mathbb{R}^{n \times n}, B \in \mathbb{R}^{n \times 1}, C \in \mathbb{R}^{1 \times n}.$$

ODE $\rightarrow$ **state-space:** for $\ddot{x} + a\dot{x} + bx = u$, let $\mathbf{x} = (x, \dot{x})^T$:

$$A = \begin{pmatrix} 0 & 1 \\ -b & -a \end{pmatrix}, B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, C = (1 \quad 0)$$

Mass-spring-damper: $m\ddot{y} + c\dot{y} + ky = u$, $G(s) = \frac{1}{ms^2 + cs + k}$.

2. Linearisation (Lec 2)

For $\dot{x} = f(x, u)$:

(1) Equilibrium $f(x_e, u_e) = 0$.

(2) Taylor (1st order):

$$\dot{x} \approx f(x_e, u_e) + \left. \frac{\partial f}{\partial x} \right|_e (x - x_e) + \left. \frac{\partial f}{\partial u} \right|_e (u - u_e)$$

(3) Deviation vars $\delta x = x - x_e$, $\delta u = u - u_e$:

$$\dot{\delta x} \approx \left. \frac{\partial f}{\partial x} \right|_e \delta x + \left. \frac{\partial f}{\partial u} \right|_e \delta u$$

Multivariable: $A = \left. \frac{\partial f}{\partial \mathbf{x}} \right|_e$ (Jacobian), $B = \left. \frac{\partial f}{\partial u} \right|_e$.

3. First-Order Systems (Lec 3)

$\dot{x} + ax = u$, free resp.: $x(t) = e^{-at}x(0)$.

Step ($u=\text{const}$), $x(0)=x_0$:

$$x(t) = \left(x_0 - \frac{u}{a}\right)e^{-at} + \frac{u}{a}$$

Steady state: $x(\infty) = u/a$. Time constant $\tau = 1/a$.

Transfer function: $y = G(s)u$, $u = Ue^{st}$.

$$G(s) = \frac{X(s)}{U(s)}. \text{ For } \ddot{x} + 3\dot{x} + 2x = u: G = \frac{1}{s^2 + 3s + 2}.$$

DC gain $\rightarrow$ **final value:** for any stable G , step of size U :

$$x(\infty) = G(0) \cdot U \quad (\text{set all derivs} = 0).$$

$G(0)$ sign = direction ($G(0)<0 \Rightarrow$ output dips below 0); magnitude = final value. Generalises $x_\infty = u/a$.

4. Proportional / Steady-State Error

Plant $\dot{x} + ax = u$, $u = K_P(r - x)$:

$$\dot{x} + (a + K_P)x = K_P r$$

$$x(\infty) = \frac{K_P r}{K_P + a}, \text{ error } e(\infty) = r \frac{a}{K_P + a}.$$

With disturbance d , noise n :

$$x_{ss} = \frac{K_P(r - n) + d}{a + K_P}$$

Large $K_P \Rightarrow$ small error (**feedback stabilisation:** pick $K_P > -a$). **Integral** action $\Rightarrow$ zero steady-state error.

5. Homogeneous Solns & Poles (Lec 4)

Char. eq. of $\ddot{x} + a\dot{x} + bx = 0$: $\lambda^2 + a\lambda + b = 0$.

Real distinct: ($a^2 - 4b > 0$): $x_h = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$.

Real repeated: ($a^2 - 4b = 0$): $x_h = (C_1 + C_2 t)e^{\lambda^* t}$.

Complex: $\lambda = -\sigma \pm j\omega_d$:

$$x_h = e^{-\sigma t}(C_3 \cos \omega_d t + C_4 \sin \omega_d t)$$

Stability: all poles (roots of denom.) have negative real part $\Leftrightarrow x(t) \rightarrow 0$ for any $x(0)$, zero input.

n th order $G(s) = \frac{b_n s^n + \dots + b_0}{s^n + a_{n-1} s^{n-1} + \dots + a_0}$; **poles**=denom roots, **zeros**=numer roots.

6. Second-Order Systems (Lec 5)

$$\text{Standard: } G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}.$$

Poles $s = -\zeta\omega_n \pm j\omega_n \sqrt{1 - \zeta^2} = -\sigma \pm j\omega_d$.

$$\sigma = \zeta\omega_n, \quad \omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\omega_n = \sqrt{\sigma^2 + \omega_d^2}, \quad \zeta = \sigma/\omega_n$$

ζ = sine of angle from imag. axis. Underdamped: $0 < \zeta < 1$.

Envelope: $x(t) = e^{-\sigma t} \sin(\omega_d t)$.

Response specifications:

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \quad (\text{overshoot})$$

$$t_p = \frac{\pi}{\omega_d} \quad (\text{peak time})$$

$$t_s \approx \frac{4}{\sigma} = \frac{4}{\zeta\omega_n} \quad (2\% \text{ settling})$$

$$t_r \approx \frac{1.8}{\omega_n} \quad (\text{rise } 10\text{--}90\%)$$

$$5\% \text{ overshoot} \Rightarrow \zeta \approx \frac{\sqrt{2}}{2} \approx 0.707.$$

7. PID Control (Lec 5)

$$C(s) = K_p + \frac{K_i}{s} + K_d s = \frac{K_d s^2 + K_p s + K_i}{s}$$

- **P:** fast, may leave SS error / oscillate.
 - **PI:** removes SS error, slower transient.
 - **PD:** adds damping, faster, noise-sensitive, adds a zero.
 - **PID:** removes SS error + improves transient.
- $K_p \uparrow \Rightarrow \omega_n \uparrow$ (faster), $\zeta \downarrow$ (more overshoot).

Rate feedback: $u = K_P(r - x) - K_D \dot{x}$ (avoids derivative kick):

$$G(s) = \frac{K_P + K_D s}{s^2 + (1 + K_D)s + K_P}$$

zero at $s = -K_P/K_D$.

PD design ex. $\ddot{x} + 3\dot{x} + 2x = u$, $u = K_p(r - x) - K_d \dot{x}$:

$$G = \frac{K_p}{s^2 + (3 + K_d)s + (K_p + 2)}$$

match to $s^2 + 2\zeta\omega_n s + \omega_n^2$.

8. Effect of Zeros (Lec 5)

$$G(s) = \frac{b_1 s + b_0}{s + a}, \text{ zero at } s = -b_0/b_1.$$

$b_1 > 0, b_0 > 0$: LHP zero, **min-phase**, kick same dir.

$b_1 < 0, b_0 > 0$: RHP zero, **non-min-phase**, "wrong-way" initial response, hard to control.

9. State-Space Response (Lec 6)

Solution: $\mathbf{x}(t) = e^{At}\mathbf{x}_0 + \int_0^t e^{A(t-\tau)}\mathbf{B}u \, d\tau$.

TF from SS: $G(s) = C(sI - A)^{-1}B + D$.

Output to $u = Ue^{st}$: $y = Ce^{At}\mathbf{x}_0 + G(s)Ue^{st}$.

Stable $\Leftrightarrow$ all eigenvalues of A have $\text{Re} < 0$.

$e^{At} = Te^{Jt}T^{-1}$ (Jordan); repeated eig give t^k terms.

10. Reachability / Control (Lec 7)

Reachability matrix:

$$W = [B \quad AB \quad A^2B \quad \dots \quad A^{n-1}B]$$

Reachable $\Leftrightarrow W$ full rank ($\det W \neq 0$ for SISO).

Equivalent: can place any closed-loop poles via K .

State feedback: $u = -K\mathbf{x}$, closed loop $\dot{\mathbf{x}} = (A - BK)\mathbf{x}$.

Design: $\det(sI - (A - BK)) \stackrel{!}{=} \text{desired char. poly}$, **match coefficients**.

Transformation: $T = W\tilde{W}^{-1}$ ($\tilde{W}$ from reachable canonical form), $K = \tilde{K}T$.

Reachable canonical form: for $\ddot{y} + a_1\dot{y} + a_2y + a_3y = u$ (top-row companion):

$$\tilde{A} = \begin{pmatrix} -a_1 & -a_2 & -a_3 \\ 1 & 0 & 0 \end{pmatrix}, \tilde{B} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

char. poly $s^3 + a_1s^2 + a_2s + a_3$ read off top row.

Dominant poles: choose 2nd-order pair for transient; place rest 5–10 $\times$ further left, low overshoot.

11. Observability / Observers (Lec 8)

Observability matrix:

$$W_o = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

Observable $\Leftrightarrow W_o$ full rank: recover $\mathbf{x}(0)$ from u, y on $[0, T]$.

Luenberger observer:

$$\dot{\hat{\mathbf{x}}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}u + \mathbf{L}(y - \mathbf{C}\hat{\mathbf{x}})$$

Error $\tilde{\mathbf{x}} = \mathbf{x} - \hat{\mathbf{x}}$: $\dot{\tilde{\mathbf{x}}} = (\mathbf{A} - \mathbf{L}\mathbf{C})\tilde{\mathbf{x}}$.

Condition: $\mathbf{A} - \mathbf{L}\mathbf{C}$ stable ($\Rightarrow \tilde{\mathbf{x}} \rightarrow 0$).

Design: $\det(sI - (\mathbf{A} - \mathbf{L}\mathbf{C})) \stackrel{!}{=} \text{desired poly}$, match coeffs.

Output feedback: (separation) observer + $u = -K\hat{\mathbf{x}} + Nr$. CL char poly: $\det(sI - (\mathbf{A} - \mathbf{B}K)) \det(sI - (\mathbf{A} - \mathbf{L}\mathbf{C}))$.

Separation principle: applies to systems that are both reachable & observable: design K and L **independently**; the CL poles are just the union of the controller poles (eig $\mathbf{A} - \mathbf{B}K$) and observer poles (eig $\mathbf{A} - \mathbf{L}\mathbf{C}$).

Feedforward: (zero SS err): $N = -(\mathbf{C}(\mathbf{A} - \mathbf{B}K)^{-1}\mathbf{B})^{-1}$.

Integral action: augment state $\dot{q} = \mathbf{C}\mathbf{x} - \bar{y}$ to kill SS error without exact model.

12. Frequency Response (Lec 9)

Input $A \sin \omega t \Rightarrow y = A|G(j\omega)| \sin(\omega t + \phi)$.

$$G(j\omega) = |G(j\omega)|e^{j\phi}, \quad \phi = \angle G = \arctan \frac{\text{Im}}{\text{Re}}.$$

Complex algebra: $z_1 z_2$: multiply mags, add phases; z_1/z_2 : divide mags, subtract phases.

$$\text{Ex. } G = \frac{1}{j\omega(j\omega + 1)(j\omega + 3)};$$

$$|G| = \frac{1}{\sqrt{\omega^2(\omega^2 + 1)(\omega^2 + 9)}}$$

$$\angle G = -\left(\frac{\pi}{2} + \arctan \omega + \arctan \frac{\omega}{3}\right)$$

Forced sinusoid (undetermined coeffs): for $\ddot{x} + \omega_n^2 x = \sin \omega t$, try $x = X \sin \omega t$, match:

$$X = \frac{1}{\omega_n^2 - \omega^2}$$

real $\Rightarrow$ phase $0^\circ/180^\circ$ (no lag); $X \rightarrow \infty$ as $\omega \rightarrow \omega_n$ (**resonance**, undamped).

13. Bode Plots (Lec 9–10)

Mag in dB: $20 \log_{10} |G(j\omega)|$. $\times 10 \Rightarrow 20\text{dB}$; $1 \Rightarrow 0\text{dB}$; $0.1 \Rightarrow -20\text{dB}$.
Asymptote rules: each **pole:** $-20\text{dB/dec} + -90^\circ$; each **zero:** $+20\text{dB/dec} + +90^\circ$ (after break freq).
Break freq at $\omega = |p|$ or $|z|$. Pure integrator $1/s$: -20dB/dec , -90° .
2nd-order: peak height & phase rate depend on ζ ; -40dB/dec , -180° .
ID from Bode: read $|G(0)|$, break points $\rightarrow$ poles/zeros, slopes; solve gain $K = |G(0)| \cdot \prod p$.

14. Nyquist Criterion (Lec 10,12)

Plot $L(j\omega)$, $\omega: -\infty \rightarrow \infty$. Closed loop $G_{yr} = \frac{L}{1+L}$.
Theorem: P =open-loop RHP poles of L ; N =net clockwise encirclements of -1 . Then CL has $Z = N + P$ RHP poles.
Stable $\Leftrightarrow Z = 0 \Leftrightarrow N = -P$ (P anticlockwise encirclements).
Stable- L case: stable iff no encirclements of -1 .
CL poles satisfy $L(s) = -1$.
Real-axis crossing: Nyquist evaluates G along the imag. axis, so to find where it crosses $(\alpha, 0)$: set $G(s)=\alpha$ and solve — the root is **purely imaginary** $s = \pm j\omega$.

15. Stability Margins (Lec 10,11)

Crossover ω_{gc} : $|L(j\omega_{gc})| = 1$ (0 dB).
Gain margin: $g_m = 1/|L(j\omega_{pc})|$ at phase $= -180^\circ$.
Phase margin: $\varphi_m = 180^\circ + \angle L(j\omega_{gc})$.
Stability margin: s_m =closest dist of Nyquist to -1 ; $|S(j\omega)| = 1/\text{dist to } (-1)$.
 $PM = 2 \sin^{-1} \left(\frac{1}{2|S(j\omega_c)|} \right)$.
Time delay: $e^{-\tau s}$: $|\cdot| = 1$, adds phase $-\omega\tau$. Max delay $\tau_{max} = \varphi_m/\omega_{gc}$.
Bode gain-phase: $\angle L(j\omega_{gc}) \approx 90^\circ \frac{d \log |L|}{d \log \omega}$; $-20\text{dB/dec} \rightarrow -90^\circ$, $-40 \rightarrow -180^\circ$.

16. Sensitivity / Gang of Four (Lec 11)

$$S = \frac{1}{1+PC}, \quad T = \frac{PC}{1+PC}, \quad S+T=1$$

Output: $\eta = Tr + PSd - Tn$; error $r - \eta = Sr - SPd + Tn$.
 $\begin{pmatrix} \eta \\ e \\ u \end{pmatrix} = \begin{pmatrix} T & PS & -T \\ S & -PS & -S \\ CS & -T & -CS \end{pmatrix} \begin{pmatrix} r \\ d \\ n \end{pmatrix}$
Disturbance: $\eta = PSd$. $|S| < 1$ attenuates; $|S|=0$ eliminates; $|S| > 1$ worsens.
 S measures sensitivity of CL to plant uncertainty.
Loop shaping — why: shape $L=PC$ to balance performance vs stability.
how: (i) high gain / integrator at low $\omega \rightarrow$ tracking + reject d ($S \approx 0$);

(ii) low gain + steep roll-off at high $\omega \rightarrow$ reject noise n ($T \approx 0$); (iii) gentle slope ($\approx -20\text{dB/dec}$) near $\omega_{gc} \rightarrow$ keep phase margin.
Or pick desired L (e.g. $\frac{\alpha}{s}$, α sets ω_{gc}) then $C=L/P$ — avoid cancelling unstable poles/zeros.

17. Fundamental Limitations (Lec 11,12)

- (1) $S + T = 1$ — seesaw trade-off.
- (2) Gain–phase relation — steep roll-off $\Rightarrow$ phase lag, low PM.
- (3) Interpolation constraints — RHP poles/zeros can't be hidden.
- (4) Bode sensitivity integral (waterbed):

$$\int_0^\infty \log |S(j\omega)| d\omega = \pi \sum_k p_k$$

p_k =unstable open-loop poles; push $|S|$ down somewhere $\Rightarrow$ up elsewhere.

18. PID Tuning — Ziegler–Nichols

Step-response (reaction curve): fit gain K , delay τ , slope; $a = K\tau/(\text{time const})$ intercept:

	k_p	T_i	T_d
P	$1/a$		
PI	$0.9/a$	$\tau/0.3$	
PID	$1.2/a$	$\tau/0.5$	0.5τ

Frequency (ultimate gain): critical gain k_c , period $T_c = 2\pi/\omega_c$:

	k_p	T_i	T_d
P	$0.5k_c$		
PI	$0.45k_c$	$T_c/1.2$	
PID	$0.6k_c$	$T_c/2$	$T_c/8$

Sluggish $\Rightarrow k_p \uparrow$ or $T_i \downarrow$; oscillatory $\Rightarrow k_p \downarrow$ or $T_d \uparrow$; SS error $\Rightarrow$ add integral.

19. Block Diagram Algebra

Series: G_1G_2 . **Parallel:** $G_1 \pm G_2$.

Neg. feedback: $\frac{G}{1+GH}$; unity: $\frac{G}{1+G}$.

Pos. feedback: $\frac{G}{1-GH}$.

$L = PC$ (loop), $G_{yr} = \frac{L}{1+L}$, $G_{er} = \frac{1}{1+L}$.

Cascaded SS: $S_1 \rightarrow S_2$ in series: get $G_i = C_i(sI - A_i)^{-1}B_i + D_i$ for each, then $G_{tot} = G_2G_1$ (multiply) — easier than building one big A .

20. Quick Reference / Exam Tips

- Stable $\Leftrightarrow$ all poles/eigenvalues in open LHP.
- Reachable: $\det[B \ AB] \neq 0$; Observable: $\det[C \ CA] \neq 0$ (2nd order).
- PD design: pick ζ from M_p , ω_n from t_s , match denom.
- Nyquist through $(\alpha, 0)$: set $G(s) = \alpha$; root is purely imaginary $s = \pm j\omega$.
- Zero SS error tracking: use **integral** action (robust) or feedforward N (needs exact model).
- Real-world uncertainty: parameter errors, disturbances, sensor noise, unmodelled nonlinearity.

PID vs state-feedback: **PID:** model-free, mainly SISO, heuristic/robust, simple to tune, limited pole control. **State-FB:** needs accurate model + full state (or observer), handles MIMO, gives exact pole placement.
Useful: $\zeta=0.5 \Rightarrow M_p \approx 16\%$; $\zeta=0.707 \Rightarrow M_p \approx 4.3\%$; $\zeta=0.6 \Rightarrow M_p \approx 9.5\%$.
 $\sqrt{1-\zeta^2}$ at $\zeta=0.707$ is 0.707.

21. Math Toolbox

Quadratic: $as^2+bs+c=0 \Rightarrow s = \frac{-b \pm \sqrt{b^2-4ca}}{2a}$.

2×2 **inverse:** $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

2×2 **eigenvalues:** $\det(\lambda I - A) = \lambda^2 - \text{tr}(A)\lambda + \det(A)$,

$$\lambda = \frac{\text{tr} \pm \sqrt{\text{tr}^2 - 4 \det}}{2}.$$

Resolvent: $(sI - A)^{-1} = \frac{\text{adj}(sI - A)}{\det(sI - A)}$.

Euler: $e^{j\theta} = \cos \theta + j \sin \theta$; $|a + jb| = \sqrt{a^2 + b^2}$, $\angle = \arctan \frac{b}{a}$.

Complex divide: $\frac{1}{a+jb} = \frac{a-jb}{a^2+b^2}$.

dB: $20 \log_{10} 2 \approx 6$, $20 \log_{10} 10 = 20$, $20 \log_{10} \frac{1}{2} \approx -6$.

Partial fractions: $\frac{1}{(s+p_1)(s+p_2)} = \frac{A}{s+p_1} + \frac{B}{s+p_2}$, $A = \frac{1}{p_2-p_1}$.

Common integrals: $\int e^{-at} dt = -\frac{1}{a} e^{-at}$; $\int_0^\infty e^{-at} dt = \frac{1}{a}$ ($a > 0$).

Trig: $\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$, $\cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$.

22. Worked Patterns

First-order ID: $x(0)=1$, $x(2)=e^{-8}$, $u=0$: $e^{-2a}=e^{-8} \Rightarrow a=4$.

Step of $\dot{x}+ax=1$, $x(0)=0$: try $x=be^{-at}+c$: $ac=1$, $b+c=0 \Rightarrow c=\frac{1}{a}$, $b=-\frac{1}{a}$. So $x=\frac{1}{a}(1-e^{-at})$.

Set $x(\infty)=X$: $ax(\infty)=u \Rightarrow u=aX$.

PD on $\ddot{x}+3\dot{x}+2x=u$: want $t_s \approx 2$, $M_p \approx 5\%$:

$\zeta = \frac{\sqrt{2}}{2}$; $t_s = \frac{4}{\zeta\omega_n} = 2 \Rightarrow \omega_n = 2\sqrt{2}$; target s^2+4s+8 .

$\Rightarrow K_d=1$, $K_p=6$.

Observer poles $-1, -2$: for $A = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$, $C=[0 \ 1]$:

$\det(sI - (A - LC)) = s^2 + (1+l_2)s + (1-l_1) \stackrel{!}{=} s^2 + 3s + 2$
 $\Rightarrow l_1 = -1$, $l_2 = 2$.

Nyquist crossing $(-0.5, 0)$: set $G(s) = -0.5$, cross-multiply, solve $\Rightarrow s^2 = -1$, $s = \pm j$.

Sensitivity, $P = \frac{1}{s(s+1)}$, $C = s+3$:

$$S = \frac{s^2+s}{s^2+2s+3}, \quad T = \frac{s+3}{s^2+2s+3}$$

Loop poles via $L(s) = -1$: solve for s ; or set $\frac{L}{1+L} = T_{des} \Rightarrow L = \frac{T_{des}}{1-T_{des}}$.